

4.2.3 Construction of Representative Group

The total area is partitioned into four sections, reflecting the four sources that influence the gasifier’s output: the overall biomass and three distinct operating parameters—temperature, equivalent air ratio, and steam to biomass ratio. To account for potential combined effects, we’ve intentionally defined each section with overlaps. To achieve optimal results, it’s crucial to iteratively refine the distributions, ensuring that deep domain knowledge about the specific processes is integrated. Below are the guiding principles used in this context:

- (a) **Significance of Parameters:** The prominence given to a particular parameter in shaping an area correlates with its importance, as drawn from domain knowledge. For instance, while temperature has its distinct section, it also serves a supplementary function in other sections.
- (b) **Attribute of Randomness:** The design ensures a degree of randomness in the distribution. By introducing randomness at an early stage, the model is fortified against high uncertainty.
- (c) **Bounds on Section Shares:** To prevent any section from becoming overly dominant, we’ve defined upper and lower limits on each section’s share. For example, we avoided omitting the SBR section to ensure consistent segmentation, avoiding fluctuation between three and four sections.
- (d) **Inclusiveness Hierarchy:** We’ve structured an internal hierarchy among the sections based on the scope of their inclusiveness.

Given the scale of our input set $[d, k, T, ER, SBR]$, we have adopted the subsequent formulation and configuration to satisfy our criteria. The coefficients were iteratively adjusted to better align with the typical composition of syngas, our target for prediction.